

# Engineering Mathematics: Analysis of Linear Systems with Parameters

Determine the values of  $\lambda$  and  $\mu$  that result in different solution types.

## 1. Problem Setup

The system of equations is:

- $x + y + z = 6$
- $x + 2y + 3z = 10$
- $x + 2y + \lambda z = \mu$

**Augmented Matrix**  $[A|B]$

$$\begin{bmatrix} 1 & 1 & 1 & 6 \\ 1 & 2 & 3 & 10 \\ 1 & 2 & \lambda & \mu \end{bmatrix}$$

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## 2. Row Transformations

We reduce the matrix to Row Echelon Form (REF) to analyze the consistency.

**Step 1: Eliminate  $x$  from  $R_2$  and  $R_3$**

$$\begin{bmatrix} 1 & 1 & 1 & 6 \\ 1 & 2 & 3 & 10 \\ 1 & 2 & \lambda & \mu \end{bmatrix} (R_2 \rightarrow R_2 - R_1, R_3 \rightarrow R_3 - R_1) \longrightarrow \begin{bmatrix} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 1 & \lambda - 1 & \mu - 6 \end{bmatrix}$$

**Step 2: Eliminate  $y$  from  $R_3$**

$$\begin{bmatrix} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 1 & \lambda - 1 & \mu - 6 \end{bmatrix} (R_3 \rightarrow R_3 - R_2) \longrightarrow \begin{bmatrix} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & \lambda - 3 & \mu - 10 \end{bmatrix}$$

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## 3. Parametric Analysis

The solution type depends on the final row:  $0x + 0y + (\lambda - 3)z = (\mu - 10)$ .

# Case 1: No Solutions

A system is inconsistent if the rank of the coefficient matrix  $A$  is less than the rank of the augmented matrix  $[A|B]$ . This happens if the last row suggests  $0 = \text{non-zero}$ .

- **Condition:**  $\lambda - 3 = 0$  AND  $\mu - 10 \neq 0$
- **Result:**  $\lambda = 3, \mu \neq 10$

# Case 2: One Solution (Unique)

A system has a unique solution if the rank of  $A$  equals the rank of  $[A|B]$  and equals the number of variables ( $n = 3$ ). This requires a non-zero pivot in every row of  $A$ .

- **Condition:**  $\lambda - 3 \neq 0$
- **Result:**  $\lambda \neq 3$  (Value of  $\mu$  can be anything)

# Case 3: Infinite Solutions

A system has infinite solutions if  $\rho(A) = \rho[A|B] < n$ . This happens if the entire last row becomes zero.

- **Condition:**  $\lambda - 3 = 0$  AND  $\mu - 10 = 0$
- **Result:**  $\lambda = 3, \mu = 10$

## 4. Summary Table

| Solution Type      | Condition on $\lambda$ | Condition on $\mu$ |
|--------------------|------------------------|--------------------|
| No Solution        | $\lambda = 3$          | $\mu \neq 10$      |
| Unique Solution    | $\lambda \neq 3$       | Any value          |
| Infinite Solutions | $\lambda = 3$          | $\mu = 10$         |